

4(a)(i)	$x_0 = 8.0 \text{ cm}$	A1
4(a)(ii)	$\omega = 2\pi / T$	C1
	$= 2\pi / 4.0 = 1.6 \text{ rad s}^{-1}$	A1
4(a)(iii)	$E = \frac{1}{2}m\omega^2x_0^2$	C1
	$= \frac{1}{2} \times 36 \times 1.6^2 \times 0.080^2$	C1
	$= 0.29 \text{ J}$	A1
4(b)	dome-shaped curve, starting and ending at $E_K = 0$	B1
	maximum E_K shown as 0.29 J	B1
	position of peak shown at $h = 10.0 \text{ cm}$	B1
	line intercepts h -axis at $h = 2.0 \text{ cm}$ and at $h = 18.0 \text{ cm}$	B1